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ESTIMATE THE WATER A DATA CENTER USES FOR COOLING

For researchers and planners comparing proposed data centers

INPUT

IT size (megawatts)
Cooling type
Watershed + month

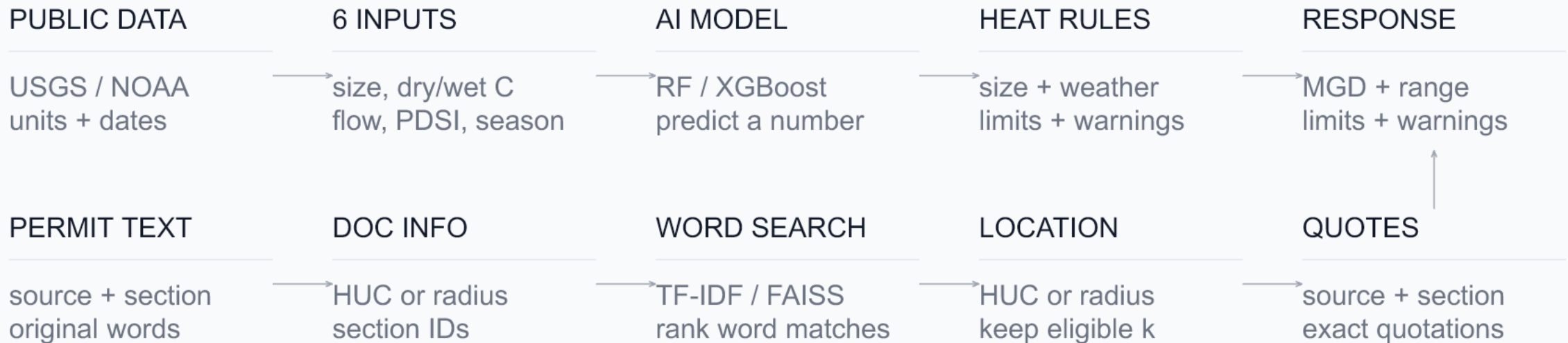
OUTPUT

Water use + physical limits
Prediction range + warnings
Permit quotes for the location

Working local prototype

TWO PATHS: PREDICT WATER USE AND FIND PERMIT QUOTES

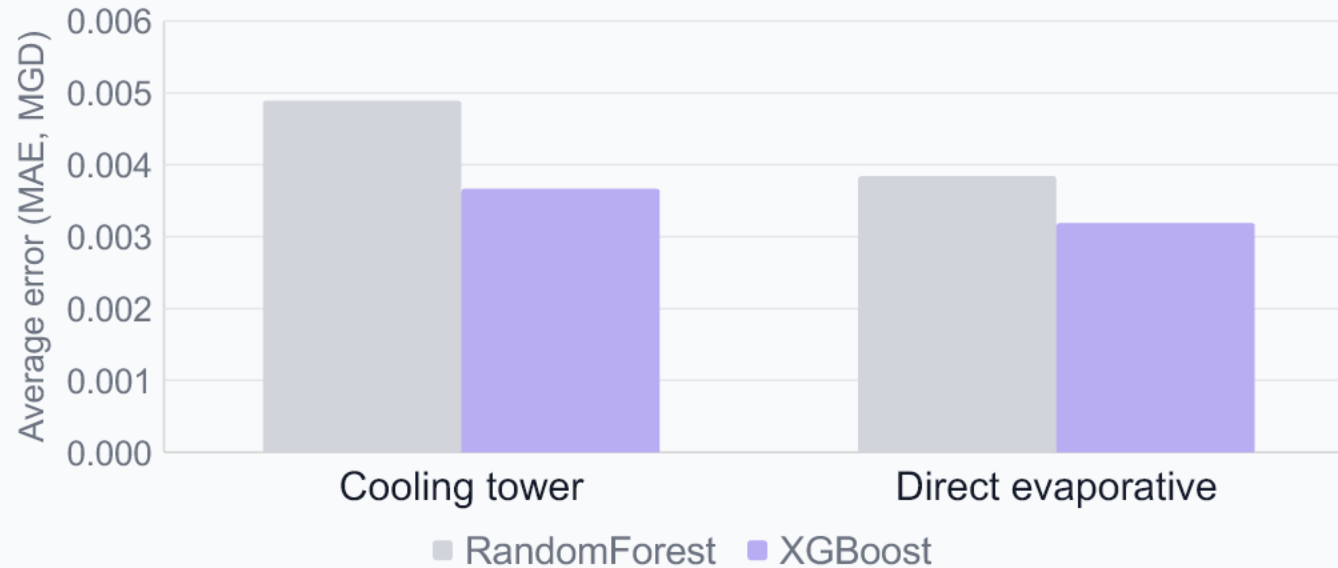
Enter IT size, cooling type, watershed code (HUC8), and month; coordinates optional



Before use: train on made-up water use; choose, set ranges, and test by date.

Forced wrong numbers trigger warnings. Four permit-location tests pass.

XGBOOST MADE SMALLER ERRORS ON OUR TEST DATA



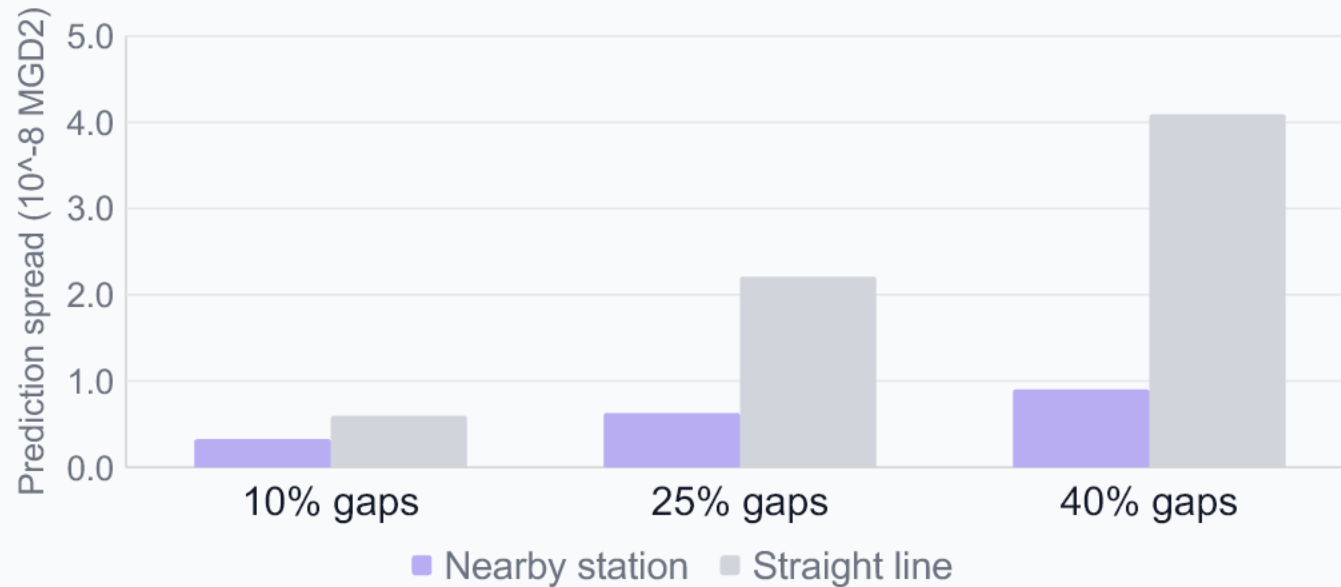
KEEP XGBOOST

Choose with RMSE on the middle date group. Keep RF for comparison.

Next: measured water use at real facilities.

Dates: train 60% / choose + set ranges 20% / test 20%
Dry air cooling: zero water target; R2 is undefined.

SIMILAR NEARBY STATIONS MADE GAPS EASIER TO FILL



100%

All main cases passed

Change within ± 0.05 MGD
50 repeats; 121 drought dates

Biased-neighbor test:
only 60.3% within limits

Improve: check neighbors and shared outages. The biased test uses a separate simple formula.

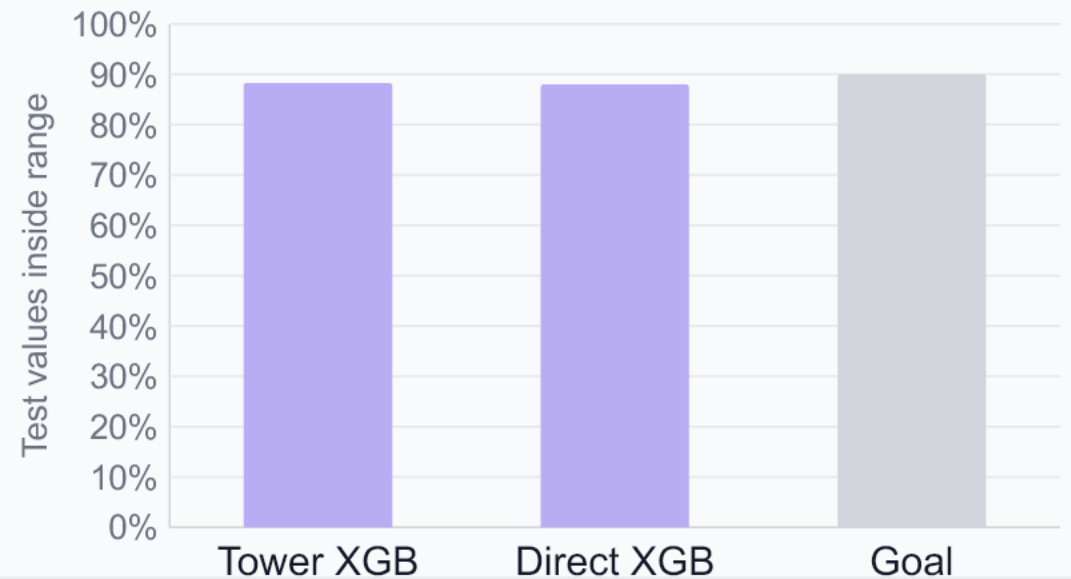
THE PREDICTION RANGES MISSED OUR 90% GOAL

88.3% / 88.0%

Tower / direct evaporative coverage

Tower R2 = 0.998

Good fit does not prove the range is reliable.



Improve: use separate data to choose models and set ranges.

Limited samples; no statistical significance test or independently verified coverage guarantee.

NEXT: TEST REAL WATER USE AT A NEW FACILITY

KEEP (ADOPT)

Tree models, physical limits, permit-location checks

IMPROVE (MODIFY)

Check neighbors, shared outages, and prediction ranges

REJECT / DEFER

R2 is not confidence; wait on neural and generative LLM models

Will it work during drought at an unseen facility, with reliable prediction ranges?

